Band-1 Hadamard order 14508 – Classical Paley-II DIRECT (q2=7253); IN SPK via Paley-II; Gap 2052 Still OPEN

H_14508 Paley-II PASS; Paley-I FAIL/SKIP (q=14507=89*163 NOT prime); Kronecker FAIL/SKIP (4*3627; 3627=3^2*13*31 odd NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-12

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-10-12_Band1_Hadamard_14504_PaleyI` (current bookkeeping); `2026-10-12_Band1_Hadamard_144` `2026-10-12_Band1_Hadamard_14476_PaleyII` (Paley-II construction + chain template); `2026-10-09_Band1_Hadamard_14460_PaleyII`; `2026-10-09_Band1_Hadamard_14432_PaleyI`; `2026-10-09_Band1_Hadamard_14428_PaleyII`; `2026-10-09_Band1_Hadamard_14356_PaleyII`; `2026-09-21_Band1_Hadamard_9124_PaleyII`; ....
Lab. `/workspace/experiments/band1/hadamard_14508_paleyII/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14508 only (verified max|HH^T-nI|=0), via classical Paley-II q2=7253 (prime ===1 mod 4), order 2(q2+1). 14508 IN SPK via Paley-II. 14508 is a long-known Hadamard order; this is a finite on-box verification, NOT a new-order claim.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14508: Paley-II **PASS** ($q_2 = 7253$ prime $\equiv 1 \pmod 4$, $n = 2(q_2+1)$); Paley-I **FAIL/SKIP** ($q = 14507 = 3 \cdot 5^2 \cdot 193$ NOT prime); Kronecker **FAIL/SKIP** ($14508 = 4 \cdot 3627$; $3627 = 7 \cdot 11 \cdot 47$ odd composite, NOT Hadamard-admissible; do NOT invent H_3627). **14508** folds into SPK **only** via classical Paley-II (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED** (no invented J). No finished-ToE claim. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14504+14480+14476+14460+14448+14432+14428+14424+14420+14412+14408+14388+14356+143** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **14508** (fifth Paley-II DIRECT wave of the 14xxx chain, after 14356, 14428, 14460 and 14476).

- H_{14508} = Paley-II ($q_2 = 7253$) – classical SPK
- Paley-I for 14508: **FAIL/SKIP** ($q = 14507 = 3 \cdot 5^2 \cdot 193$ NOT prime)
- Kronecker for 14508: **FAIL/SKIP** ($4 \cdot 3627$; $3627 = 7 \cdot 11 \cdot 47$ odd composite, NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14508.py`

Results. overall_pass=True; units 852/852; row blocks 114/114 (block size 128); wall 7472.5675s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 14508 = 2(q_2 + 1)$ with $q_2 = 7253$ prime $\equiv 1 \pmod 4$ — Paley-II eligible (claim path).
- Paley-I: $q = n - 1 = 14507 = 3 \cdot 5^2 \cdot 193$ NOT prime — FAIL/SKIP.
- Kronecker: $14508 = 4 \cdot 3627$; $3627 = 7 \cdot 11 \cdot 47$ odd composite ($\equiv 3 \pmod 4$) NOT Hadamard-admissible — FAIL/SKIP; no other admissible split.
- Shape: $14508 = 2^2 \cdot 7 \cdot 11 \cdot 47 \equiv 12 \pmod{16}$ (same ===12 mod 16 odd-composite-m class as template $14476 = 2^2 \cdot 3 \cdot 5 \cdot 241$; prior twin-class $14028 = 2^2 \cdot 3 \cdot 7 \cdot 167$ unbuilt).
- Miyamoto digression skipped this wave (prefer classical SPK).

Construction (Paley type II)

Let χ be the quadratic character of $\mathbb{F}_{7253}$ and $Q_{ij} = \chi(j - i)$ (symmetric, since $q_2 \equiv 1 \pmod 4$). The conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & Q \end{pmatrix}$ of order $q_2 + 1 = 7254$ gives

$$H_{14508} = \begin{pmatrix} C + I & C - I \\ C - I & -(C + I) \end{pmatrix}.$$

Built directly in int8 (memory-lean; unit-tested equal to the int64 reference builder on $q \in \{5, 13, 17, 29, 37, 41, 53, 61\}$), then verified block-wise: every row block of $HH^T - nI$ checked exactly in int32.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14508 Paley-II	Paley-II q2=7253	0	7453.145
H_14508 Paley-I	FAIL/SKIP (q=14507=89*163 NOT prime)	–	–
H_14508 Kronecker	FAIL/SKIP (*43627; 3627=3ˆ2*13*31 odd NOT Hadamard-admissible)	–	–

SPK closure note

Quantity	Value
14508 in SPK	**True** (Paley-II q2=7253)
Prior Paley-I 14504	IN SPK via Paley-I q=14503 (~6:50PM ET 2026-10-12)
Prior Paley-I 14480	IN SPK via Paley-I q=14479 (~5:00PM ET 2026-10-12)
Prior Paley-II 14476	IN SPK via Paley-II q2=7237 (~3:02PM ET 2026-10-12)
Prior Paley-II 14460	IN SPK via Paley-II q2=7229 (~9:30PM ET 2026-10-09)
Prior 14448	IN SPK via Paley-I (~8:47PM ET 2026-10-09)
Prior 14432	IN SPK via Paley-I (~7:51PM ET 2026-10-09)
Prior Paley-II 14428	IN SPK via Paley-II q2=7213 (~7:10PM ET 2026-10-09)
Prior Paley-II 14356	IN SPK via Paley-II q2=7177 (~1:22PM ET 2026-10-09)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false

- GATE-04 LOCKED (no invented J)
- No finished-ToE claim
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- 14508 is a long-known order (Paley 1933 family); finite on-box verification only
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

NEXT direct route after 14508 (FIRST any direct = Paley-I): q=14519->n=14520 Paley-I DIRECT (prime ===3 mod 4; Paley-II q2=7259=7*17*61 NOT prime (===3 mod 4) -> FAIL/SKIP; Kronecker 14520=4*3630; 3630===2 mod 4 NOT Hadamard-admissible -> FAIL/SKIP; template hadamard_14504_paleyI / hadamard_14480_paleyI) | no Kronecker-only in 14512..14516 | skipped Paley-II DIRECT orders below 14508 still unbuilt (template hadamard_14508_paleyII / hadamard_14476_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)